

Across-Array LDPC Codes Design for Resistive Random-Access Memories

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Abstract—Resistive random-access memory (ReRAM) has garnered significant attention due to its high storage density, rapid read/write speeds, and compatibility with CMOS devices. However, the simple crossbar structure of ReRAM introduces sneak path (SP) interference, which impairs the storage reliability of the ReRAM system. To address this, we propose a novel zoned namespace (ZNS)-based ReRAM storage scheme, which is well-suited for scenarios with hybrid storage protection of multitype data and employs low-density parity-check (LDPC) codes to mitigate the impact of SPs. Furthermore, considering the asymmetric nature of the ReRAM channel, we first develop a computer-calculable asymmetric discrete density evolution of belief propagation decoding (ADDE-BP). Due to the across-array storage structure, SP interference and noise vary across subarrays, resulting in codewords experiencing a nonuniform “sum channel.” To address this, we further propose criteria based on ADDE-BP for nonuniform error correction, specifically for the design of LDPC codewords. Simulation results show that the proposed codes achieve 1–2 orders of magnitude lower bit error rate (BER) than IEEE standard codes and Mackay codes under most channel conditions, thus demonstrating that our proposed LDPC code design for across-array applications enhances data storage reliability compared to conventional methods. Most importantly, the maximum check node degree and node type of the designed code were reduced, thus greatly simplifying the hardware implementation. This work establishes a co-design framework for ReRAM that simultaneously improves the reliability and efficiency of storage systems.

Index Terms—Across-array data storage, density evolution, low-density parity-check (LDPC) codes, nonuniform error correction, zoned namespace (ZNS)-based resistive random-access memory (ReRAM).

I. INTRODUCTION

IN THE era of big data, reliable data storage cannot be ignored. To date, the most widely used nonvolatile memory is flash memory [1], [2], which boasts faster read/write performance and higher storage density than mechanical hard disks. However, as the storage density of flash memory devices continues to increase, it is subject to certain limitations [3]. For instance, the programming voltage in the flash memory cannot be reduced below a certain level, leading to serious

programming errors. In addition, the internal charge is more prone to leakage, compromising the reliability of user data. In recent years, resistive random-access memory (ReRAM) [4] has risen as a notable contender to flash memory, offering distinct advantages across multiple domains. The theoretical foundation for memristors, derived from the term “memory resistor,” was laid by Chua in 1971 [5]. Memristors distinguish different logical data through variations in resistance values; the high-resistance (HR) state represents a logical “0,” while the low-resistance (LR) state represents a logical “1.” Now, memristors have emerged as fundamental components for implementing in-memory computing (IMC) paradigms [6], effectively addressing the critical “memory wall” bottleneck. The distinctive plasticity characteristics of memristors provide a novel physical foundation for developing energy-efficient computing systems that integrate storage and computation [7]. This unique capability enables the design of memory-centric architectures where data processing occurs directly within memory units, significantly reducing energy consumption and latency associated with traditional von Neumann architectures.

The crossbar array structure [8] is commonly employed in ReRAM for its benefits, such as high-density storage and efficient and rapid data access. The memories are located at the intersection of the columns and rows of the array. During read operations, a specific voltage is applied to the target cell, and its resistance is inferred by measuring the current that flows through the memory. Eventually, the logical data is determined through the hard-decision process based on the resistance value. However, this straightforward array structure and reading mechanism also introduce a significant challenge known as sneak path (SP) interference [9], as depicted in Fig. 1. The SP is influenced by the data distribution and the array size. Typically, the retrieved current is not only from the target cell but represents the “sum current” in a parallel circuit configuration, suggesting that the combined resistance is lower than that of any single branch, as represented by $R_{\text{sum}} < \min\{R_1, R_2, \dots, R_n\}$. This can lead to an underestimation of the target resistance, potentially causing erroneous data detection.

In the existing literature, numerous works have dedicated efforts to enhancing the reliability of ReRAM through various avenues, including material properties, channel modeling, noise detection, and the development of channel coding and error correction techniques. Cassuto et al. [10] proposed a method to address the SP effect using constrained codes. This approach restricts stored data to specific regions at the cost of reduced code rate. In terms of ReRAM fabrication, the incorporation of a selector device ensures unidirectional current

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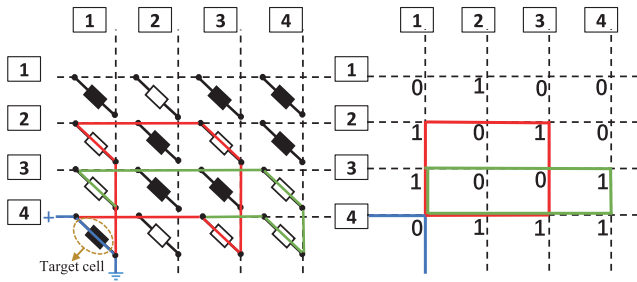


Fig. 1. Illustration of 4×4 memory array of ReRAM with SP (left). The corresponding data array (right).

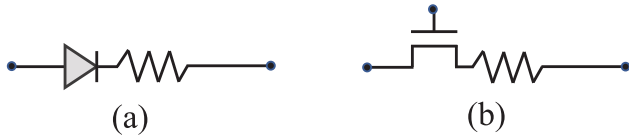


Fig. 2. (a) Corresponds to the “1D1R” structure, which employs the diode as the selector. (b) Shows the structure of “1T1R,” adopting the transistor-MOS as the selector.

flow, thereby diminishing the likelihood of SP occurrences. As demonstrated in Fig. 1, the current directions in the target and SPs exhibit opposite polarity. To establish unidirectional current flow during read operations, the integration of a selector device during fabrication emerges as an essential strategy. The slight reduction in storage density is negligible compared to the significant improvement in mitigating interference in the SP.

In current memristor architectures, these selector configurations have been widely adopted: 1-diode-1-resistor (1D1R) and 1-transistor-1-resistor (1T1R), and their structures are shown in Fig. 2.

As illustrated in Fig. 1, SP interference is intrinsically pattern-dependent. Chen et al. [11] identified a significant correlation among interference events affecting elements in the same rows and columns. To leverage this correlation during data detection, they proposed a pilot-cell-based storage paradigm, wherein adaptive threshold detection schemes utilize side information from these pilot cells. While this approach enhanced data reliability in architectures without error correction coding (ECC), it inevitably compromised storage efficiency due to the overhead of pilot cell allocation. Concurrently, with the rapid advancement of deep learning, [12] introduced a detection architecture based on deep neural networks (DNNs). This approach accomplishes data detection with minimal channel modeling information; however, the generalization capability of such models requires further validation across diverse operational conditions and array configurations.

In light of the remarkable success achieved by low-density parity-check (LDPC) codes in improving data reliability within flash memory [13], [14], the ReRAM also adopts LDPC codes [15] to mitigate SP interference. In [16], a two-step bits allocation strategy has been proposed. This strategy disperses multiple codeword bits across distinct storage arrays based on the node relationships within the Tanner graph. Although

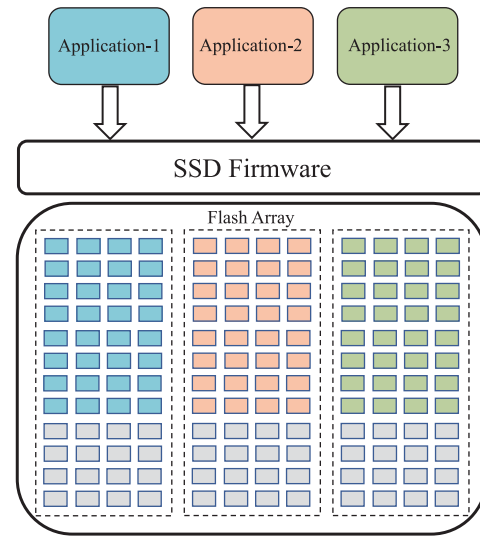


Fig. 3. ZNS architecture, which organizes data based on application-specific and data-type criteria.

this method diminishes the influence of channel correlation within a single codeword, it does not address the interference between different codewords. Song et al. [17] introduced an across-array storage scheme aimed at reducing channel correlation between bits by distributing codewords across multiple memory arrays, demonstrating that these bits are subject to independent channels. However, this approach focused solely on decoding optimization and differs significantly from our research.

Recently, solid-state drive (SSD) manufacturers have introduced a novel data placement mechanism known as zoned namespace (ZNS) [18], which organizes data based on application-specific and data-type criteria, as shown in Fig. 3. The ZNS architecture, contrasting with conventional block-interface SSDs, effectively alleviates issues about flash throughput degradation, diminished lifespans, and the expansion of the over-provisioning (OP) space. Furthermore, ZNS demonstrates significant potential for hybrid storage systems managing diverse data types, a prevalent requirement in modern big data applications. The proven success of ZNS on SSDs motivates its adaptation to ReRAM, a promising candidate for next-generation nonvolatile memory. This study introduces a methodology for designing LDPC codes specifically optimized for a ZNS-based ReRAM architecture, where across-array data storage is completed depending on the data type. Song et al. [17] observed that across-array data storage in ReRAM reduces data correlation, potentially enhancing the effectiveness of traditional ECCs. However, when a single codeword spans multiple arrays, it may be susceptible to nonuniform interference patterns, which could compromise the ECC's performance. The reasons are as follows: 1) Generally, data types, such as text, images, video, system data, and databases, exhibit different distributions for binary bits 0's and 1's [19]. 2) As mentioned earlier, the SP interference encountered by the data in ReRAM is intricately linked to the data distribution within the array and the random interference exhibiting variation between subarrays. Thus, the optimization

of ECC for across-array storage is necessary. In this article, the LDPC code is considered. Conventional, regular LDPC codes exhibit limited resilience to nonuniform interference. While irregular LDPC codes offer improved resistance, a systematic design methodology specifically targeting the characteristics of nonuniform interference in this context remains an open research problem. To address these limitations, we propose an asymmetric discrete density evolution method for the nonuniform errors (ADDE-BP-NON) method, aimed at crafting LDPC codes that enhance performance and ensure the reliability of stored data. Computer-aided design tools developed by the above theoretical methods are effective in generating such good codewords. To this end, the contributions of our work are as follows.

- 1) For the hybrid storage scenarios of multitype data prevalent in the big data era, we propose a novel ZNS-based ReRAM storage architecture, where the architecture allocates the different types of data into multiple sub-arrays and employs a single hybrid LDPC code to protect them. Furthermore, recognizing the asymmetric and nonuniform interference characteristics inherent in this multitype data storage scenario, we introduce a tailored across-array LDPC code design methodology to enhance storage reliability.
- 2) For the asymmetric channel, we propose a computer-calculable belief propagation (BP)-based asymmetric density evolution (ADE) algorithm. Although we only address the ReRAM channel in this article, the algorithm inherently applies to the progressivity analysis of the codeword in any asymmetric channel.
- 3) Furthermore, an asymmetric discrete density evolution method is proposed for nonuniform errors. Compared to conventional density evolution techniques, the resulting codeword design offers superior error correction performance and reduced hardware implementation complexity. Notably, this work represents the first instance of code design tailored explicitly for asymmetric channels exhibiting nonuniform error characteristics.

The article is organized as follows. Section II provides fundamentals for the ReRAM system. Section III gives the storage architecture of ZNS-based ReRAM and two types of LDPC code design methods, asymmetric discrete density evolution of BP decoding (ADDE-BP) and ADDE-BP-NON. Code design and results are presented in Section IV. Finally, Section V concludes our contributions.

II. RESISTIVE MEMORY CHANNEL MODEL

We examine an $m \times n$ memory array of ReRAM, where the memristor at the intersection of the i th row and the j th column is denoted as a cell (i, j) , with $i \in \{1, 2, \dots, m\}$ and $j \in \{1, 2, \dots, n\}$. Meanwhile, $x_{i,j} \in \{0, 1\}$ represents an element of the data matrix X , and $y_{i,j}$ corresponds to the output signal upon reading. In the data writing process, a specific voltage is applied to modulate the resistance of the target cell according to the hysteresis characteristic curve of the memristor [5]. Conversely, during the data erasure phase, all cells are uniformly transitioned to the HR state. Despite

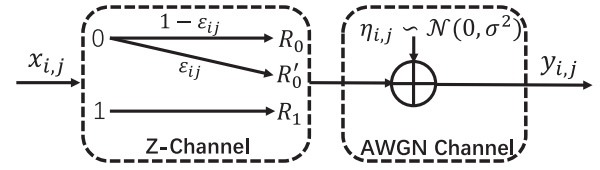


Fig. 4. ReRAM channel model, where R_1 indicates LR value, R_0 and R'_0 denotes HR value with/without SP interference. We take the resistance values of $R_0 = 1000 \, \Omega$, $R_1 = 100 \, \Omega$ and $R'_0 = 250 \, \Omega$ [17].

the simplicity of ReRAM's operational mechanism, its storage performance is significantly compromised by SP interference. SPs constitute unintended current pathways that run parallel to the expected current path, and these paths are considered interference. Fig. 1 illustrates SP interference, where cell $(4, 1)$ is the target cell under reading. The intended current path is indicated by the blue line, contrasted by two SPs: 1) the red path $(4, 1) \rightarrow (4, 3) \rightarrow (2, 3) \rightarrow (2, 1) \rightarrow (4, 1)$ and 2) the green path $(4, 1) \rightarrow (4, 4) \rightarrow (3, 4) \rightarrow (3, 1) \rightarrow (4, 1)$. The SP interference adversely impacts HR cells, thus rendering it difficult to clearly distinguish between HR cells and LR cells during the read phase. It is important to note that the current in the SP flows in the opposite direction to the intended current. So, a selector is integrated to ensure that the current can only flow in one direction, thus inhibiting the emergence of the SP. However, it is prone to failure due to manufacturing defects or material degradation, which can potentially reactivate the SP.

The formation of a SP can be attributed to these primary factors: 1) the specific storage patterns, the target cell $x_{i,j} = 0$ and $x_{i,s} = x_{k,s} = x_{k,j} = 1$, $k \in \{1, 2, \dots, m\}$, $s \in \{1, 2, \dots, n\}$; and 2) the failure of the selector device associated with the cell (k, s) . Here, we focus predominantly on the SP of length-3 for facilitating analysis. The longer SP in series exhibits greater resistance compared to those of length-3 and consequently has a diminished effect on the measured resistance of the target cell. Meanwhile, the number of SPs under consideration is typically modest, given the low likelihood of concurrent selector failures across multiple cells.

The ReRAM channel is characterized by a unique Z-channel configuration in tandem with an additive Gaussian noise channel, as depicted in Fig. 4. We utilize additive Gaussian noise to represent the random noise [20] encountered by the resistances and the measurement noise introduced by the sensing circuit, consistent with the approach in [17]. However, our methodology can be expanded to include other composite noise models, as indicated in [16]. According to the channel model, the asymmetric nature of the ReRAM channel is defined by two parameters: (ϵ_{ij}, σ) . The term ϵ_{ij} denotes the probability that a SP occurs within the cell (i, j) , associated with the input data distribution and the specific cell. The standard deviation σ characterizes resistance variations in the array, where uniform deviation distribution across memory cells in a single array is conventionally assumed.

By the conditions that govern the emergence of SP interference, we can conduct a quantitative analysis of the average probability of the occurrence of SP interference within an array of size $m \times n$. We presuppose the input distribution to be independent and identically distributed (i.i.d.) Bernoulli

process with parameter q , i.e., $P(x_{i,j} = 1) = q$ and $P(x_{i,j} = 0) = 1 - q$. Then, a binary random variable $e_{i,j}$ is introduced to characterize whether cell (i, j) is impacted by SP interference, where $e_{i,j} = 1$ indicates that cell (i, j) is contaminated (i.e., $Pr(e_{i,j} = 1 | x_{i,j} = 0) = \epsilon_{ij}$) and $e_{i,j} = 0$ indicates it is not (i.e., $Pr(e_{i,j} = 0 | x_{i,j} = 0) = 1 - \epsilon_{ij}$). Therefore, the resistance of the readout can be formulated as follows:

$$r_{i,j} = \begin{cases} R_1, & x_{i,j} = 1 \\ \frac{1}{\frac{1}{R_0} + \frac{\epsilon_{i,j}}{R_s}}, & x_{i,j} = 0 \end{cases} \quad (1)$$

where R_s is paralleled resistance value caused by SPs interference with the target cell and $R'_0 = ((1/R_0) + (1/R_s))^{-1}$.

According to the principle of SP formation, the HR cell will not encounter SP interference when the following two points are satisfied simultaneously: 1) Certain positions in the same row or column are HR cells, and then these paths cannot form the SPs (contrary to condition 2 of the principle of SP formation). 2) The remaining positions in the same row or column are LR cells. Still, their intersection positions are the LR cells whose selectors are not disabled or HR cells (characterized by probability $1 - p_f q$). These paths cannot form the SPs (contrary to condition 3 of the principle of SP formation). Therefore, for a specified distribution ($q = q^*$), the ϵ_{q^*} (mean value of ϵ_{ij}) can be calculated by

$$\begin{aligned} \epsilon_{q^*} &= E \left[\frac{\sum_{i=1}^m \sum_{j=1}^n e_{ij}}{mn(1 - q^*)} \right] \\ &= 1 - \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} \binom{m-1}{i} \binom{n-1}{j} \times \\ &\quad q^{*(i+j)} (1 - q^*)^{(m-1-i+n-1-j)} (1 - p_f q^*)^{ij}. \end{aligned} \quad (2)$$

The p_f is the failure probability of the selector, and it is set to 0.001 [16], [17]. For convenience in subsequent calculations, the factor $f(q) = (1 - p_f q)^{ij}$ in (2) can be simplified by Taylor expansion with the Peano remainder term. To minimize the simplification error, the Taylor expansion can be applied here. In practice, the tolerable error E is considered in advance, and the order (n) of the expansion is determined

$$\frac{f^{(n)}(q_0)}{n!} (q - q_0)^n \leq E. \quad (3)$$

From (2), it shows that the larger the array size, the more likely the SPs will occur. Moreover, in the hybrid storage of multitype data, the data suffers from nonuniform errors due to the different information distributions and the different specifications of the stored arrays. So, it is more challenging to protect the data in ReRAM to fulfill both high-capacity and high-density requirements for the hybrid storage of multitype data.

III. ACROSS-ARRAY LDPC CODING DESIGN

The probability of the SPs' occurrence depends on the distribution of the stored data and the specifications of the storage array. To achieve high reliability and high-density storage, we implement the ZNS-based [18] storage architecture for ReRAM, where the across-array data storage [17] is based

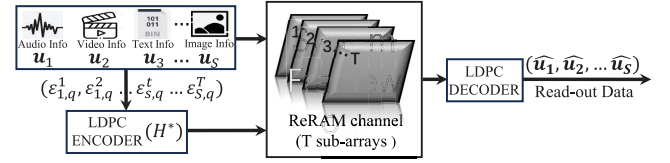


Fig. 5. System model for hybrid LDPC coded for ZNS-based ReRAM storage.

on the data type. The schematic of the storage architecture is presented in Fig. 5. In this scenario, we examine a typical case where a single LDPC codeword is allocated to multiple subarrays for storage.

In this study, we presume the existence of S distinct types of data. Considering source bits marked by S sets of binary data vector of length- k_s : $\mathbf{u}_s = (u_1, u_2, \dots, u_{k_s})$, $s \in \{1, 2, \dots, S\}$, these bits are allocated to T subarrays. The likelihood of SP occurrences within each subarray is denoted as $\epsilon^t_{s,q}$, $t \in \{1, 2, \dots, T\}$, $s \in \{1, 2, \dots, S\}$ and is contingent upon the distribution of stored data. Following [17], we substitute $\epsilon^t_{s,q}$ with its mean value $\epsilon^t_{s,q}$, which is calculated using (2). In this storage paradigm, based on (6) of [17], the ReRAM storage channel can be equated to a block-varying channel. And the channel parameters $\epsilon^t_{s,q}$ may vary among subarrays. Additionally, the noise (σ_t) is not uniform across different subarrays. Consequently, the “sum channel” experienced by the LDPC code constitutes a nonuniform channel within ReRAM. Given the asymmetrical nature of the ReRAM channel, conventional LDPC code design methodologies are inadequate to address these issues in tandem. Drawing from [21], [22], we have constructed an ADDE-BP-NON in this section.

A. Asymmetric Discrete Density Evolution for BP

Given that the ADE algorithm, as introduced by Wang et al. [22], lacks explicit expressions for iterative BP decoding, is practical intractable for its high complexity. To solve this problem, we develop a computer-calculable iterative formula for the ADDE-BP.

The uniform quantizer $W(x)$ [23] is considered

$$W(x) = \begin{cases} \lceil \frac{x}{\Delta} - \frac{1}{2} \rceil \cdot \Delta, & x \leq -\frac{\Delta}{2} \\ \lfloor \frac{x}{\Delta} + \frac{1}{2} \rfloor \cdot \Delta, & x \geq \frac{\Delta}{2} \\ 0, & \text{others} \end{cases} \quad (4)$$

where x and $W(x)$ represent the message prior to and following quantization, respectively; The symbol Δ denotes the quantization interval; The operations $\lceil x \rceil$ and $\lfloor x \rfloor$ correspond to the ceiling and floor functions, respectively. In practice, we first determine the quantization range by simulating the maximum value of the log-likelihood ratios (LLR), and then take the smallest possible quantization interval while considering the complexity, thus preventing channel loss.

In BP decoding, the LLR of the channel output serves as the decoding input. The output of the variable nodes (VNs) and the check nodes (CNs) are mapped according to (5) and

(6), respectively

$$W(v_i^{(l)}) = W(v_i^{(0)}) + \sum_{j=1}^{d_v-1} W(u_j^{(l)}) \quad (5)$$

$$\tanh \frac{W(u_j^{(l)})}{2} = \prod_{i=1}^{d_c-1} \tanh \frac{W(v_i^{(l-1)})}{2} \quad (6)$$

where $v_i^{(l)}$ represents the message transmitted from i th variable node (VN_i) at the l th iteration and d_v is the degree of this variable node. Similarly, $u_j^{(l)}$ signifies the message sent from the j th check node (CN_j) at the l th iteration, and d_c is the degree of this check node.

To simplify the notation, discrete messages are redefined as follows: $W(v_i^{(l)}) \Rightarrow \hat{v}_i^{(l)}$; $W(u_j^{(l)}) \Rightarrow \hat{u}_j^{(l)}$. Under *independence assumption* [21], the individual messages $\hat{v}_i^{(l)}$ and $\hat{u}_j^{(l)}$ in the l th iteration are considered random variables of i.i.d. Consequently, the subscripts in $\hat{v}_i^{(l)}$ and $\hat{u}_j^{(l)}$ can be removed, denoting them as $\hat{v}^{(l)}$ and $\hat{u}^{(l)}$, respectively. For the ADE approach, it is necessary to analyze the evolution of both 0-code and 1-code messages throughout the iterative process. Therefore, $P_z^{(l)}(\hat{v})$ is defined as the density function of the message emanating from VN at the l th iteration, with the subscript $z \in \{0, 1\}$ representing the case of 0-code and 1-code. Additionally, $P_z^{(0)}(\hat{v})$ ($l = 0$) signifies the initial message density of the channel. Likewise, $Q_0^{(l)}(\hat{u})$ and $Q_1^{(l)}(\hat{u})$ denote the density function of the message of CN .

1) Message Density Update of Variable Node:

$$P_z^{(l)}(\hat{v}) = P_z^{(0)}(\hat{v}) \otimes Q_z^{(l)}(\hat{u})^{\otimes (d_v-1)}. \quad (7)$$

Here, the computation of message density of VN is consistent with the original density evolution [23]. In practical implementation, the discrete sequence convolution can be accelerated with the fast Fourier transform (FFT) and the inverse FFT (IFFT).

2) Message Density Update of Check Node:

$$R_2(a, b) = 2 \tanh^{-1} \left(\tanh \frac{a}{2} \times \tanh \frac{b}{2} \right). \quad (8)$$

For the sake of simplicity, we first introduce the R_2 operator (8) introduced in [23]. Then, (9) employs the R_2 operator to rerepresent (6), where vector $\hat{v} = \{\hat{v}_1, \hat{v}_2, \dots, \hat{v}_{d_c-1}\}$. And, in order to make the expression of the density evolution formula of the check nodes more convenient and concise, the function $\Phi_{(c)}(\cdot)$ is used to express the update process of the check node

$$\begin{aligned} \hat{u} &= \Phi_{(c)}(\hat{v}) = 2 \cdot \tanh^{-1} \left(\prod_{i=1}^{d_c-1} \tanh \frac{\hat{v}_i}{2} \right) \\ &= R_2(\hat{v}_1, R_2(\hat{v}_2, \dots, R_2(\hat{v}_{d_c-2}, \hat{v}_{d_c-1}))). \end{aligned} \quad (9)$$

The derivation of the check node evolution formula plays a key role in ADDE-BP. The Tanner graph of a check node with its neighboring variable nodes during the iterative process is given in Fig. 6. Since each check node is i.i.d., the output message density of a certain node is denoted by $Q_z^{(l)}(u)$, $z \in \{0, 1\}$. According to the concept of “perfect projection” [22], it suffices to provide the message densities $Q_0^{(l)}(u)$ and $Q_1^{(l)}(u)$

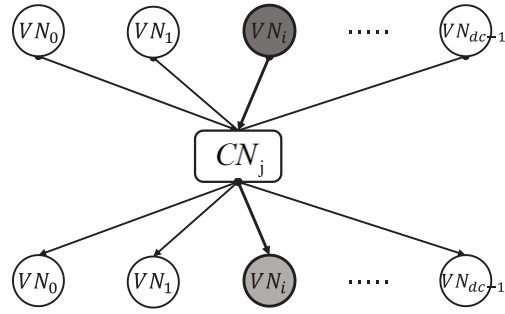


Fig. 6. Partial exhibition of the Tanner graph.

that correspond to the nodes VN_i receiving decision values of 0 and 1, i.e., $Q_0^{(l)}(u) \mapsto V_i = 0$; $Q_1^{(l)}(u) \mapsto V_i = 1$. Furthermore, we define a neighborhood set of CN_j as $N_{CN_j}^1 = \{VN_0, VN_1, \dots, VN_{d_c-1}\}$. Given that the decision values of the nodes within $N_{CN_j}^1$ must satisfy $\text{mod}(\sum_{i=0}^{d_c-1} V_i, 2) = 0$, when $V_i = 0$ is ascertained, it follows that $\sum_{k=1, k \neq i}^{d_c-1} V_k = n$ must be even and its probability is

$$P(n = \text{even} | V_i = 0) = \binom{d_c-1}{n} (P(V_k = 1))^{d_c-2}. \quad (10)$$

Similarly, when $V_i = 1$

$$P(n = \text{odd} | V_i = 1) = \binom{d_c-1}{n} ((P(V_k = 1)))^{d_c-2}. \quad (11)$$

Finally, using (10) and (11), the updated formula for CN 's message density can be given

$$Q_0^{(l)}(\hat{u}) = \frac{1}{2^{d_c-2}} \sum_{k=0, k=\text{even}}^{d_c-1} \binom{d_c-1}{k} \cdot \sum_{\hat{v}: \Phi_{(c)}(\hat{v})=\hat{u}} \prod_{d=1}^k P_1^{(l-1)}(\hat{v}_d) \cdot \prod_{d=k+1}^{d_c-1} P_0^{(l-1)}(\hat{v}_d) \quad (12)$$

$$Q_1^{(l)}(\hat{u}) = \frac{1}{2^{d_c-2}} \sum_{k=0, k=\text{odd}}^{d_c-1} \binom{d_c-1}{k} \cdot \sum_{\hat{v}: \Phi_{(c)}(\hat{v})=\hat{u}} \prod_{d=1}^k P_1^{(l-1)}(\hat{v}_d) \cdot \prod_{d=k+1}^{d_c-1} P_0^{(l-1)}(\hat{v}_d). \quad (13)$$

In particular, when $k = 0$ in (12), $\prod_{d=1}^k P_1^{(l-1)}(\hat{v}_d) = 1$, and when $k = d_c - 1$ in (13), $\prod_{d=k+1}^{d_c-1} P_0^{(l-1)}(\hat{v}_d) = 1$.

B. ADDE-BP for Nonuniform Errors

The ADDE-BP-NON method proposed in this section is motivated by the potential to enhance the performance of irregular LDPC codes through degree distribution optimization. From the perspective of nonuniform error immunity, due to the different number of CN s connected to different VN s of irregular LDPC codes, the information received by each VN during the decoding process is different, which leads to VN s possessing nonuniform immunity to interference. However, in the traditional design of irregular codes, nonuniform immunity is not considered as a priori metric. Inspired by this, designing LDPC codes with nonuniform immunity corresponding to

known nonuniform interference is possible. Furthermore, the “sum channel” in the ZNS-based storage architecture also has asymmetric properties. Therefore, we propose the ADDE-BP-NON method to approach the optimal matching between LDPC codes and nonuniform asymmetric channels.

In contrast to ADDE-BP, the ADDE-BP-NON categorizes VNs based on the interference intensity they experienced (i.e., subchannels). Then, distinct “subdegree distributions” are designed for each category, eventually forming the complete degree distribution. Specifically, during the calculation of the VN probability density, the probability density function (PDF) for each subchannel is first derived independently. Subsequently, the PDFs of the “sum channel” are combined using a weighted average, where the weights correspond to the proportion of edges in the Tanner graph associated with each subchannel category relative to the total number of edges. Due to the analytical complexity of nonuniform CNs degree distribution matching over the channel, current optimization efforts for CNs distribution are aligned with conventional irregular code design principles. While ADDE-BP can be conceptualized as a single optimization problem for code design under a channel, ADDE-BP-NON can be considered a joint optimization problem, seeking an optimal code design for the effective channel formed by the aggregation of multiple subchannels, each with distinct interference characteristics. Critically, ADDE-BP-NON accounts for the heterogeneous interference profiles across subchannels during density evolution. By tailoring the degree distribution irregularity to each subchannel, the resulting composite code, when applied to the overall aggregated channel, exhibits a highly effective degree distribution, leading to enhanced error correction performance.

For the convenient presentation of the algorithm, it is assumed that a length- n codeword is allocated to T subarrays during data writing, and the subsequent notations are hereby defined.

- 1) n_t denotes the number of bits stored (transmitted) on the t th subarray (subchannel). And $\sum_{t=1}^T n_t = n$.
- 2) $\eta_t = \{x_{t1}, x_{t2}, x_{t3}, \dots\}$, where index $t_1, t_2, t_3, \dots \in \{1, 2, \dots, n\}$, denotes the set of bits stored (transmitted) on the t th subarray (subchannel). So $|\eta_t| = n_t$, where $|\cdot|$ denotes the cardinality of the set. And let $G_t = (n_t/n)$.
- 3) E is the set of edges in the Tanner graph of the LDPC codes. E_t represents the set of edges connected with a variable node of type t , and the corresponding bits are stored (transmitted) in the t th subarray (subchannel). Furthermore, E_t^i represents the set of edges attached to the variable node with the degree i and the t th subarray.
- 4) $\lambda_{(t)}(x) = \sum_i \lambda_{(t)}^i \cdot x^{i-1}$ is defined as the degree distribution of the variable node under t th subchannel, where $\lambda_{(t)}^i = (|E_t^i|/|E|)$. The definition of check nodes is similar. To facilitate the design of codewords, the degree distribution of the check nodes is maintained consistent with the traditional definition [23].
- 5) $\Lambda(x) = \{\lambda_{(1)}(x), \lambda_{(2)}(x), \dots, \lambda_{(T)}(x)\}$ and $\rho(x) = \sum_i \rho_i x^{i-1}$ represent the degree distribution of variable nodes and check nodes of the LDPC code that can correct nonuniform errors within a codeword. The ensemble $C(\Lambda, \rho)$ is the collection of LDPC codes.

Algorithm 1: ADDE-BP-NON

Input: Channel parameters: $(\epsilon_q^1, \epsilon_q^2, \dots, \epsilon_q^T)$ and $(\sigma_1, \sigma_2, \dots, \sigma_T)$.
 R_2 operator pre-lookup table: R_{LUT} .
 VNs degree distribution: $(\lambda_{(1)}(x), \lambda_{(2)}(x), \dots, \lambda_{(T)}(x))$.
 CNs degree distribution: $\rho(x)$.
 Vector of bits ratio over sub-array: $(G_1, G_2, \dots, G_T)$.

```

1 Initialization  $l = 0, l_{\max}, W(x)$  and  $stop\_pe$ .
2 for  $t \in [1, T]$  do
3    $(\epsilon_q^t, \sigma_k) \Rightarrow P_{z,t}^{(0)}(\hat{v})$ ;
4    $(G_t, \lambda_{(t)}(x), E) \Rightarrow Pr_t$ ;
5 end
6 The initial density function of the “sum channel”:
   $P_z^{(0)}(\hat{v}) = \sum_{t=1}^T Pr_t * P_{z,t}^{(0)}(\hat{v})$ .
7 Iterative Process:
8 while  $l < l_{\max} \cap pe > stop\_pe$  do
9   CN update:
10   $Q_z^{(l)}(\hat{u}) = \psi_{even/odd}(P_0^{(l-1)}(\hat{v}), P_1^{(l-1)}(\hat{v}), \rho(x), R_{LUT})$ .
11  VN update:
12  for  $t \in [1, T]$  do
13     $P_{z,t}^{(l)}(\hat{v}) = P_{z,t}^{(0)}(\hat{v}) \otimes \lambda_{(t)}(Q_z^{(l)}(\hat{u}))$ ;
14  end
15  The updated density function of the “sum channel”:
   $P_z^{(l)}(\hat{v}) = \sum_{t=1}^T Pr_t * P_{z,t}^{(l)}(\hat{v})$ .
16  Calculation of error probability:  $P_e^{(l)}$ .
17  Update  $l \leftarrow l + 1$ .
18 end
Output:  $P_e^{(l)}$  and Convergence or not?

```

- 6) $Pr_t = (|E_t|/|E|)$ is the ratio of the number of edges associated with the variable node in the t th subarray to the number of all edges in the Tanner graph. Essentially, $Pr_t, t \in \{1, 2, \dots, T\}$ determines the irregularity of the codewords in the ensemble $C(\Lambda, \rho)$, while this is governed by the channel nonuniformity.

The proposed ADDE-BP-NON is described in Algorithm 1, which can examine the convergence of an LDPC code set with the VNs degree distribution $\{\lambda_{(1)}(x), \lambda_{(2)}(x), \dots, \lambda_{(T)}(x)\}$ and the CNs degree distribution $\rho(x)$ over a nonuniform and asymmetric channel characterized by parameters $(\epsilon_q^1, \epsilon_q^2, \dots, \epsilon_q^T)$ (omitting the subscript s) and $(\sigma_1, \sigma_2, \dots, \sigma_T)$. Additionally, the prelookup table (R_{LUT}) of the R_2 operator and the vector of bits ratio over subarray $(G_1, G_2, \dots, G_T)$ serve as inputs. During the initialization of the algorithm, the maximum number of iterations ($l_{\max}$), the quantizer parameters ($W(x)$), and the stopping error rate ($stop_pe$) are provided. In step 3, $P_{z,t}^{(0)}, z \in \{0, 1\}, t \in \{0, 1, \dots, T\}$ denotes one of the T pairs of initial LLR message density functions. And $P_z^{(0)}, z \in \{0, 1\}$ is the initial density function of the “sum channel”.

The iterative process of Algorithm 1 includes the following phases.

1) *Message Density Update of Check Node:* $Q_z^l, z \in \{0, 1\}$ means the same as with ADDE-BP. For the sake of illustration, two transformation operators for the check nodes, denoted ψ_{odd} and ψ_{even} , are utilized to represent (12) and (13), respectively. Specifically, Q_0^l is obtained by mapping ψ_{even} , and ψ_{odd} to Q_1^l .

2) *Message Density Update of Variable Node:* The variable nodes are updated following the conventional DE approach.

However, the distinction is that the update of variable nodes must be differentiated by subchannel and calculated individually.

Algorithm 1 measures the threshold of the LDPC codes in an asymmetric and nonuniform channel, where the threshold level is representative of the performance of the codes to some extent. In particular, a pair of channel parameters (ϵ, σ) is employed to characterize the threshold of codes during density evolution. In the next section, we apply the ADDE-BP-NON method and the differential evolution method to design the LDPC codes for the ReRAM channel. To simplify the comparison of thresholds during differential evolution, the smallest interference intensity $(\epsilon_{\min}, \sigma_{\min})$ among the nonuniform interferences is considered as the threshold for density evolution.

IV. CODES DESIGN AND RESULTS

In this section, to verify the validity of the across-array storage scheme for hybrid storage of multitype data, a bunch of “good” LDPC codes are designed using a computer-aided design tool developed by the ADDE-BP-NON methodology. The thresholding of the ensemble of LDPC codes can be determined by the density evolution, and it is necessary to find the pair of degree distributions that produces the maximum possible noise threshold for a given channel. The problem is a nonlinear cost function minimization problem with continuous space parameters. A commonly used solution method is the differential evolution algorithm [24]. This algorithm has been successfully applied to the design of LDPC codes for erasure channels [25], AWGN channels [21], and Rayleigh channels [26]. Differential evolution encompasses four primary stages: initialization, mutation, crossover, and selection. Following the crossover operation, which generates candidate degree distributions, the ADDE-BP-NON method is employed to evaluate the thresholds of each resulting degree distribution. This threshold evaluation serves as the basis for the subsequent selection stage. Generally, with the increase of T , the parameter search space for code design becomes excessively broad, increasing the complexity of the design. The matrix (14) represents the parameters of the variable nodes

$$\begin{bmatrix} \lambda_{1,1} & \lambda_{1,2} & \cdots & \lambda_{1,d_{v1}} \\ \lambda_{2,1} & \lambda_{2,2} & \cdots & \lambda_{2,d_{v2}} \\ \cdot & \cdot & \cdots & \cdot \\ \lambda_{t,1} & \lambda_{t,2} & \cdots & \lambda_{t,d_{vt}} \\ \cdot & \cdot & \cdots & \cdot \\ \lambda_{T,1} & \lambda_{T,2} & \cdots & \lambda_{T,d_{vT}} \end{bmatrix}. \quad (14)$$

Considering the tradeoff between design complexity and code performance, the “pseudo-regular codes” within the ensemble $\mathcal{C}(\Lambda, \rho)$ serve as an alternative, characterized by $\lambda_{(i)}(x) = x^{d_{vi}-1}$, $t \in \{1, 2, \dots, T\}$, where the parameters matrix has a single nonzero element per row. In essence, the degree distribution of the variable nodes is focused on a single degree within each subchannel, albeit with some inevitable performance degradation. To substantiate the effectiveness of our designed codewords, we compare their performance with the IEEE standard code for flash memory [27] and the Mackay LDPC code [28], as detailed in Table I.

TABLE I
DETAILED PARAMETERS OF LDPC CODES

Parameters	n	λ	ρ	R
IEEE Standard Code	10080	$\lambda_3 = 0.0105$ $\lambda_4 = 0.9895$	$\rho_{23} = 0.0802$ $\rho_{24} = 0.9198$	0.83
Mackay Code	4095	$\lambda_4 = 1$	$\rho_{22} = 0.8048$ $\rho_{23} = 0.1952$	0.82

TABLE II
EXAMPLE: DEGREE DISTRIBUTION FOR $T = 4$ AND $R = 0.8$

Parameters	λ	ρ	threshold
Pseudo regular	$d_{v1} = 3$ $d_{v2} = 3$ $d_{v3} = 6$ $d_{v4} = 3$	$\rho_{19} = 1$	$\sigma_{\min} = 45$
Fully irregular	$\lambda_{1,3} = 0.1214, \lambda_{1,6} = 0.2008$ $\lambda_{2,3} = 0.2219$ $\lambda_{3,3} = 0.203, \lambda_{3,5} = 0.0312$ $\lambda_{4,3} = 0.2217$	$\rho_{16} = 0.092$ $\rho_{17} = 0.908$	$\sigma_{\min} = 46$

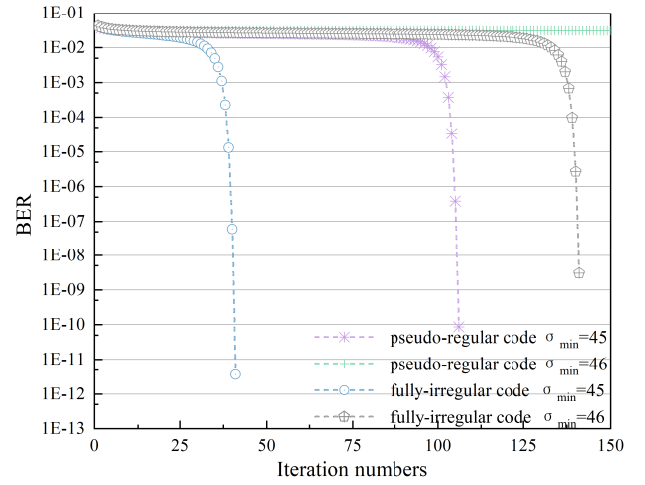


Fig. 7. Asymptotic BERs of the example over the ReRAM channels ($T = 4$) predicted by the ADDE-BP-NON, where just the green line indicates the case of nonconvergence. Because the channel condition ($\sigma_{\min}$) exceeds the threshold for pseudo-regular code.

Table II are designed for a ZNS-based ReRAM architecture adopting the across-array storage scheme, where $T = 4$ and the channel parameters are $\{\epsilon_q = (0.5, 0.4, 0.4, 0.6), \sigma = (\sigma_{\min} + 10, \sigma_{\min} + 5, \sigma_{\min} + 5, \sigma_{\min})\}$. The $\sigma_{\min}$ is used as a measure of the codeword threshold. It can be seen that the threshold of “fully irregular code” is better than that of the “pseudo-regular code.” Furthermore, the maximum degree of “fully irregular code” is smaller than “pseudo-regular code,” IEEE standard code and Mackay code, which is more beneficial to hardware implementation.

Fig. 7 illustrates the asymptotic bit error rates (BERs) of the example over the ReRAM channels ($T = 4$). It is observed that fully irregular codes exhibit more rapid convergence. These codes require fewer iterations under identical channel conditions, as demonstrated by the comparison with pseudo-regular codes and fully irregular codes at $\sigma_{\min} = 45$ in Fig. 7. Concurrently, once the channel noise exceeds the threshold for

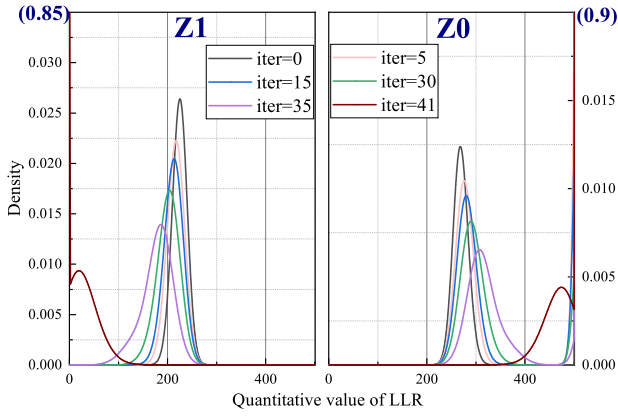


Fig. 8. Evolution of the density function of “fully irregular code” under $\sigma_{\min} = 45$, all-ones code (left) and all-zeros code (right). The log-likelihood ratio (LLR) quantization parameters: $\{-50:0.2:50\}$.

the codes, the BERs do not exhibit a sustainable decrease, as exemplified by the pseudo-regular code at $\sigma_{\min} = 46$.

In contrast to conventional density evolution methods, the ReRAM channel exhibits inherent asymmetry, necessitating the consideration of both the all-zeros and all-ones density functions in the design and analysis of codes. Fig. 8 illustrates the evolution of the density function for a fully irregular code at $\sigma_{\min} = 45$. The right-hand side of Fig. 8 depicts the LLR evolution of the all-zeros code, demonstrating that the LLR distribution shifts toward positive values as the iteration count escalates. Conversely, the LLR distribution of the all-ones code veers toward the negative values. Moreover, the probability of error can be progressively minimized, fulfilling the convergence criterion $P_e^{(l)} \leq \text{stop } p_e$.

It can be seen in Fig. 9 that the performance of the designed “pseudo-regular code” and “fully irregular code” with code length $n = 4096$ outperforms the other two codes. Specifically, for most of the channel noise ($\sigma_{\min}$), the BERs of fully irregular code are about 1-order of magnitude lower than that of the pseudo-regular code. Compared to Mackay LDPC code, the BERs of fully irregular code can be reduced by at least 2-order of magnitude. Meanwhile, the maximum node degree of “fully irregular code” is lower than that of “pseudo-regular code,” while the decoding performance is better.

Subsequently, we examine the performance degradation of the LDPC codeword designed using $\epsilon_{s,q}^t$ under actual channel conditions. As shown in Fig. 10, the codeword, designed with the average value $\epsilon_{s,q}^t$, experiences a marginal loss in performance in the real channel. Furthermore, as described in [17], the adoption of the across-array scheme results in the channel parameters ϵ_s^t becoming increasingly centered around its mean value with an increasing number of across-array T . Consequently, this leads to a further reduction in performance loss.

To demonstrate the merits of the across-array LDPC coding design in a ZNS-based ReRAM storage architecture, we take a set of source information with three types of data as the storage pattern, where the corresponding parameters are: $q_1 = 0.55$, $n_1 = 4096$, $q_2 = 0.6$, $n_2 = 1024$ and $q_3 = 0.75$, $n_3 = 1024$, respectively. Fig. 11 compares the performance adopting

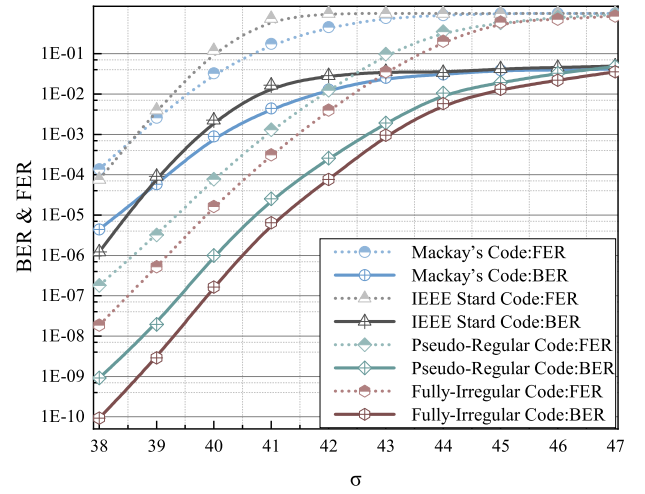


Fig. 9. Performance comparison between different codes under the same channel conditions: $\{\epsilon_q = (0.5, 0.4, 0.4, 0.6), \sigma = (\sigma_{\min} + 10, \sigma_{\min} + 5, \sigma_{\min} + 5, \sigma_{\min})\}$.

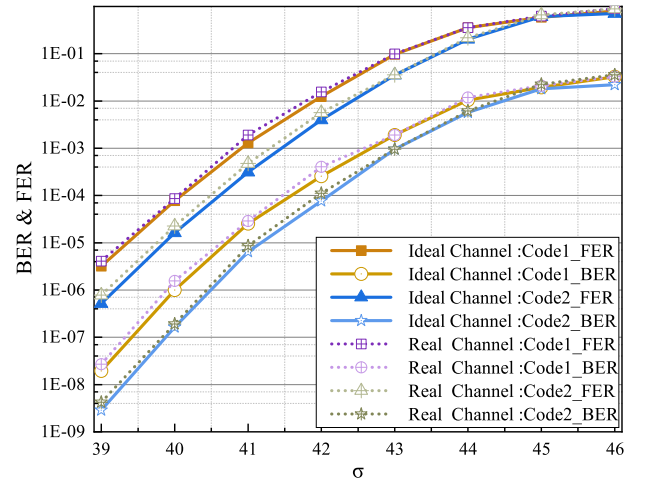


Fig. 10. Performance comparison of code for the ideal channel and real channel. The ϵ_s^t of real channel is varying around $\epsilon_{s,q}^t$. The degree distributions of code 1 and code 2 are the same as the pseudo-regular code and the fully irregular code listed in Table II.

different numbers of across-arrays ($T = 1, 3, 4$). To ensure fair comparisons, we clarify the configurations for different numbers of across-arrays: $T = 1$ indicates that three types of data are stored individually and jointly protected using a single hybrid code. Channel parameters are averaged across subchannels. Thus, the code design does not account for nonuniform interference. $T = 3$ represents each of the three data types stored in a dedicated subarray. A single hybrid code provides joint error protection. Crucially, the code design explicitly considers the individual channel conditions of each subarray. $T = 4$ denotes that the first type of data is stored in two subarrays and also protected together with the remaining two types of data using one single hybrid code (similar to the $T = 3$ case). The code rate ($R = 0.87$) is fixed for all T . The channel parameters of different across-array numbers are listed here, where the parameters characterizing the SP interference ($\epsilon_{s,q}^t$) are computed using (2).

TABLE III
DETAILED PARAMETERS OF LDPC CODES UNDER DIFFERENT ACROSS-ARRAY SCHEMES

Parameters	T	λ	ρ
Uniform Channel	$T = 1$	$\lambda_2 = 0.1186, \lambda_3 = 0.366, \lambda_4 = 0.1668$ $\lambda_5 = 0.1866, \lambda_6 = 0.152$	$\rho_{27} = 0.1858, \rho_{28} = 0.8142$
	$T = 3$	$\lambda_{1,3} = 0.5217, \lambda_{2,5} = 0.2174, \lambda_{3,6} = 0.2609$	$\rho_{29} = 1$
Non-uniform Channel	$T = 4$	$\lambda_{1,3} = 0.2609, \lambda_{2,3} = 0.2609, \lambda_{3,5} = 0.2174$ $\lambda_{4,6} = 0.2608$	$\rho_{29} = 1$

- 1) Uniform channel $T = 1$, the joint channel parameters of the three arrays:

$$\epsilon_q = (\epsilon_{1,q}^1 + \epsilon_{2,q}^2 + \epsilon_{3,q}^3)/3, \quad \sigma = (\sigma_1 + \sigma_2 + \sigma_3)/3.$$

- 2) Nonuniform channel $T = 3$, the joint channel parameters of the three arrays:

$$\epsilon_{1,q}^1 = 0.48, \quad \sigma_1 = \sigma_{\min} + 8.$$

$$\epsilon_{2,q}^2 = 0.1864, \quad \sigma_2 = \sigma_{\min} + 5.$$

$$\epsilon_{3,q}^3 = 0.3322, \quad \sigma_3 = \sigma_{\min}.$$

- 3) Nonuniform channel $T = 4$, the joint channel parameters of the four arrays:

$$\epsilon_{1,q}^1 = 0.2760, \quad \sigma_1 = \sigma_{\min} + 8.$$

$$\epsilon_{2,q}^2 = 0.2760, \quad \sigma_1 = \sigma_{\min} + 8.$$

$$\epsilon_{3,q}^3 = 0.1864, \quad \sigma_2 = \sigma_{\min} + 5.$$

$$\epsilon_{3,q}^4 = 0.3322, \quad \sigma_3 = \sigma_{\min}.$$

For the above three storage scenarios (channel conditions), we utilize the proposed algorithm for the corresponding LDPC code design with the parameters listed in Table III, where the core optimization procedure is consistent with the standard differential evolution algorithm. The performance comparison is shown in Fig. 11, and we have several remarks.

- 1) Compared to $T = 1$ (without the across-array scheme), the storage reliability can be dramatically improved after adopting the across-array scheme with the proposed code design algorithms (ADDE-BP-NON algorithms). The reason is that our method takes into account the effect of nonuniform interference on the probability density of VNs in the probability density tracking so that the complete codeword can balance the anti-noise performance of each type of VNs. Ultimately, the nonuniform interference immunity of the codeword is improved.
- 2) When choosing a higher across-arrays number, a larger diversity gain can be obtained in terms of reliability, which is consistent with the analysis of [17].
- 3) Using a limited variety of variable node types, we have achieved an enhancement in storage performance, which significantly reduces the complexity of hardware implementation, thereby further validating the practicality and feasibility of our proposed scheme.

The above three points demonstrate that the proposed code design methodology in the ZNS-based ReRAM storage scheme can effectively resist data-related SP interference and array-related random electronic interference. Furthermore, it has a greater advantage regarding complexity than traditional design methods.

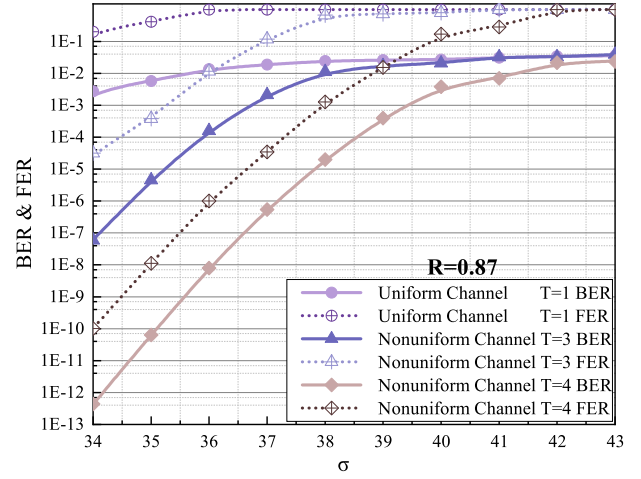


Fig. 11. Performance comparison for the ReRAM channel with adopting different numbers of across-array. The channel condition is determined by the fixed source we choose and the corresponding size of the subarray.

V. CONCLUSION

This article introduces a novel ZNS-based ReRAM storage architecture for scenarios with hybrid storage protection of multitype data, which leverages data type segregation for across-array data placement. This architecture inherently mitigates the interference correlation between stored data, thereby enhancing the effectiveness of ECC schemes. Further, to efficiently resolve asymmetric and nonuniform errors arising from two primary sources: 1) SP interference and 2) inherent device variability. We propose a novel across-array LDPC code design methodology based on a computer-calculable asymmetric discrete density evolution tailored for nonuniform error channels. This approach enables the construction of LDPC codes specifically optimized to resist the asymmetric and nonuniform interference characteristic of ReRAM. Consequently, LDPC codes designed using this method demonstrate superior performance in ensuring ReRAM storage reliability. Furthermore, compared to conventional LDPC design techniques, the resulting codes exhibit a significantly reduced maximum check node degree and fewer variable node types, highlighting their advantages in terms of design efficiency and simplified hardware implementation. Importantly, the proposed LDPC code design can be applied to other communication and storage channels exhibiting asymmetric and nonuniform error characteristics straightforwardly.

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